ABSTRACT

University hostel accommodation in Kenya has faced ongoing challenges. Statistics show that 65% of students are unhappy with their living arrangements, and over 40% request room changes during their first semester. Traditional systems have relied mainly on manual processes or simple first-come-first-served methods. These often ignore important factors like student preferences, lifestyle compatibility, academic schedules, and past patterns. This has resulted in poor roommate matches, troubled living situations, underused facilities, and increased administrative workload.

The main goal of this project was to create a Smart Room Allocation System that uses machine learning and mobile technology to improve hostel room assignments. The study followed an Agile methodology, which allowed for gradual development and continuous feedback. The system was built on a three-tier architecture. It features a React Native mobile application for the frontend, a Node.js backend with RESTful APIs, and a Python-based machine learning engine using Scikit-learn for the recommendation logic.

Key features of the system include digital student profiles, a strong room inventory management module, and a hybrid recommendation algorithm. This algorithm mixes content-based filtering with rule-based logic to calculate compatibility scores between students and available rooms.

The results of the implementation showed significant improvement over traditional methods. System testing found a 92% accuracy rate in matching students based on their stated preferences, such as sleeping habits, study hours, and social tendencies. User Acceptance Testing (UAT) indicated an 89% satisfaction rate with the usability of the mobile interface. This project clearly demonstrates that including Artificial Intelligence in university administrative processes can streamline operations and greatly enhance the student living experience.

Keywords: Room Allocation, Machine Learning, Mobile Application, Compatibility Matching, React Native, Scikit-learn.